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# MAST30027 2021 special special exam

Modern Applied Statistics (University of Melbourne)



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THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Special on Special Assessment, 2021

School of Mathematics and Statistics

# MAST30027 Modern Applied Statistics

Reading time: 30 minutes — Writing time: 2 hours — Upload time: 30 minutes

This exam consists of 14 pages (including this page) with 5 questions and 50 total marks

## Permitted Materials

- This exam and/or an offline electronic PDF reader, one or more copies of the masked exam template made available earlier, blank loose-leaf paper and a Casio FX-82 calculator.
- One double sided A4 page of notes (handwritten only).
- No headphones or earphones are permitted.

## Instructions to Students

- Wave your hand right in front of your webcam if you wish to communicate with the supervisor at any time (before, during or after the exam).
- You must not be out of webcam view at any time without supervisor permission.
- You must not write your answers on an iPad or other electronic device.
- Off-line PDF readers (i) must have the screen visible in Zoom; (ii) must only be used to read exam questions (do not access other software or files); (iii) must be set in flight mode or have both internet and Bluetooth disabled as soon as the exam paper is downloaded.

*Writing*

- If you are writing answers on the exam or masked exam and need more space, use blank paper. Note this in the answer box, so the marker knows.
- If you are only writing on blank A4 paper, the first page must contain only your student number, subject code and subject name. Write on one side of each sheet only. Start each question on a new page and include the question number at the top of each page.

*Scanning and Submitting*

- **You must not leave Zoom supervision to scan your exam.** Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned exam as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary. Do not leave Zoom supervision until you have confirmed orally with the supervisor that you have received the Gradescope confirmation email.
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**Question 1 (7 marks)**

The inverse Gaussian distribution has p.d.f.

$$f(x;\mu,\lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} \exp\left(\frac{-\lambda(x-\mu)^2}{2\mu^2 x}\right)$$

for $x > 0$, where $\mu > 0$ and $\lambda > 0$.

(a) Show that the inverse Gaussian distribution is an exponential family. [hint: you could consider $\theta = -1/\mu^2$.]



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(b) Obtain the canonical link and the variance function.

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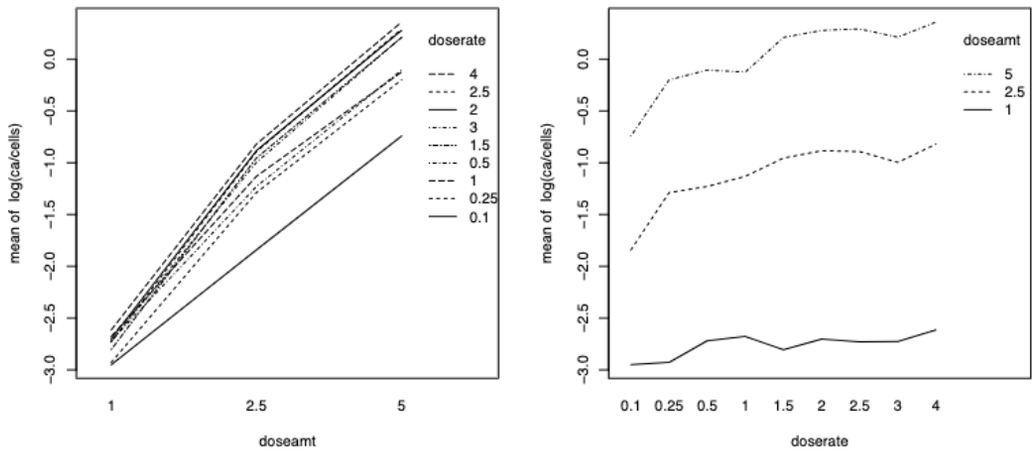
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**Question 2 (8 marks)**

In Purott and Reeder (1976), some data is presented from an experiment conducted to determine the effect of gamma radiation on the numbers of chromosomal abnormalities (ca) observed. The observed counts ranged from 25 to 419. The number (cells), in hundreds of cells exposed in each run, differs. The dose amount (doseamt) and the rate (doserate) at which the dose is applied are the predictors of interest.

```
> library(faraway)
> data(dicentric)
> par(mfrow=c(1,2))
> with(dicentric, interaction.plot(doseamt, doserate, log(ca/cells),
+ lty=c(1,2,4,5,6,1,2,4,5)))
> with(dicentric, interaction.plot(doserate, doseamt, log(ca/cells), lty=c(1,2,4)))
> par(mfrow=c(1,1))
```




Some R output from an analysis of these data follows. Use it to answer the questions that follow.

```
> model <- glm(ca ~ offset(log(cells)) + doserate*doseamt, family=poisson, data=dicentric)
> summary(model)
Call:
glm(formula = ca ~ offset(log(cells)) + doserate * doseamt, family = poisson,
    data = dicentric)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-5.7308  -2.2842  -0.6264   3.3487   5.8272

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)      -3.29994    0.06160 -53.567  < 2e-16 ***
doserate          0.06401    0.02922   2.191 0.028476 *
doseamt           0.61224    0.01707  35.862  < 2e-16 ***
doserate:doseamt  0.02715    0.00765   3.549 0.000387 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```



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```
(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 4753.00  on 26  degrees of freedom
Residual deviance:  270.26  on 23  degrees of freedom
AIC: 453.67

Number of Fisher Scoring iterations: 4

> anova(model, test="Chi")
Analysis of Deviance Table

Model: poisson, link: log

Response: ca

Terms added sequentially (first to last)


                Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
NULL                                26     4753.0
doserate         1    231.3        25     4521.7 < 2.2e-16 ***
doseamt          1   4238.7        24      282.9 < 2.2e-16 ***
doserate:doseamt  1     12.7        23      270.3 0.0003681 ***
---
Signif. codes:  0 ’***’ 0.001 ’**’ 0.01 ’*’ 0.05 ’.’ 0.1 ’ ’ 1
```

(a) Specify the model fitted.

(b) What evidence do you have to support the linear assumptions in the model?

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(c) The significance of the interaction term was tested twice, giving $p$-values of 0.000387 and 0.0003681. Define the test statistic used in each case, and the distributions used to obtain the $p$-values. (You may use terms such as deviance and Fisher information without giving their exact formula.)

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**Question 3 (10 marks)**

We assume that $X_1, \ldots, X_n$ are independent to each other, and each $X_i$ follows a mixture of two binomial distributions. Specifically, for $i = 1, \ldots, n$,

$$Z_i \sim \text{categorical } (\pi, 1-\pi),$$

$$X_i | Z_i = 1 \sim \text{Binomial}(20, p_1) \text{ and } X_i | Z_i = 2 \sim \text{Binomial}(20, p_2).$$

We aim to obtain the MLE of the parameters $\theta = (\pi, p_1, p_2)$ using the EM algorithm. The binomial distribution has probability mass function

$$f(x; m, p) = \binom{m}{x} p^x (1-p)^{m-x}.$$

Note: If your final answers contain $P(Z_i = 1 | X_i, \theta^0)$ or $P(Z_i = 2 | X_i, \theta^0)$, you do not need to provide detailed formulae for them, i.e., your final answers can contain these expressions.

(a) Let $X = (X_1, \ldots, X_n)$ and $Z = (Z_1, \ldots, Z_n)$. Derive the expectation of the complete log-likelihood, $Q(\theta, \theta^0) = E_{Z|X,\theta^0}[\log(P(X, Z|\theta))]$.



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(b) Derive the M-step of the EM algorithm.

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**Question 4 (13 marks)**

**Model:** Consider a normal model:

$$x \sim \text{Normal}(\mu, \sigma^2).$$

We observe $x = 0$.

**Note 1:** The exponential distribution with rate $\gamma > 0$ (i.e., Exponential(rate = $\gamma$)) has the probability density function

$$f(x; \gamma) = \gamma e^{-\gamma x}$$

for $x \geq 0$.

**Note 2:** The Normal distribution with mean $\mu$ and variance $\sigma^2$ (i.e., $\text{N}(\mu, \sigma^2)$) has the probability density function

$$f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

**Note 3:** The Inverse-Gamma($a$, $b$) has the probability density function

$$f(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp\left(-\frac{b}{x}\right).$$

(a) **Prior 1:** We impose the following prior for the mean and variance parameters:

$$p(\mu, \sigma^2) \propto \frac{1}{\sigma^2}$$

**Metropolis-Hastings algorithm:** we wish to perform posterior inference using the Metropolis-Hastings algorithm. For the current values of the parameters $(\mu_c, \sigma_c^2)$, we propose new values $(\mu_n, \sigma_n^2)$ as follows: $\sigma_n^2 \sim$ Exponential(rate = $\frac{1}{\sigma_c^2}$) and $\mu_n \sim$ Normal(mean = $\mu_c$, variance = $\sigma_n^2$). Compute the acceptance probability when $(\mu_c, \sigma_c^2) = (-1, 2^2), (\mu_n, \sigma_n^2) = (1, 1^2)$. Show your work.

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(b) **Prior 2:** We impose the following prior for the mean and variance parameters:

$$p(\mu, \sigma^2) = p(\mu|\sigma^2)p(\sigma^2),$$

$$\mu|\sigma^2 \sim \text{Normal}(1, \sigma^2), \quad \sigma^2 \sim \text{Inverse-Gamma}(1, 1),$$

**Variational Inference (VI):** we wish to apply VI with the mean-field variational family where $q(\mu, \sigma^2) = q_\mu(\mu)q_{\sigma^2}(\sigma^2)$, and use the CAVI algorithm for optimisation. The CAVI iteratively optimises each factor as follows while holding the other factors fixed:

$$\begin{aligned} q^*_\mu(\mu) &\propto \exp\{\mathbb{E}_{\sigma^2}[\log p(\mu, \sigma^2, x)]\}, \\ q^*_{\sigma^2}(\sigma^2) &\propto \exp\{\mathbb{E}_\mu[\log p(\mu, \sigma^2, x)]\}, \end{aligned}$$

where the expectations $\mathbb{E}_{\sigma^2}$ and $\mathbb{E}_\mu$ are taken with respect to $q^*_{\sigma^2}(\sigma^2)$ and $q^*_\mu(\mu)$, respectively. Derive $q^*_{\sigma^2}(\sigma^2)$ and identify the corresponding distribution name and its parameters. The parameters may contain $\mathbb{E}_\mu[\mu]$ or $\mathbb{V}_\mu[\mu]$, which are the mean and variance of $q^*_\mu(\mu)$, respectively.

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**Question 5 (12 marks)**

Let $X \sim \mathrm{N}(\mu, 1^2)$. The p.d.f. for the normal distribution is given in Question 1. We impose a prior on $\mu$ as follows:

$$\begin{cases} \mu \sim \mathrm{N}(1, 1^2) & \text{if } \gamma = 1, \\ \mu = 0 & \text{if } \gamma = 0, \end{cases}$$

and

$$\begin{cases} \gamma = 1 & \text{with probability } 0.8, \\ \gamma = 0 & \text{otherwise.} \end{cases}$$

We observe $X = 2$. Write down an algorithm to simulate 100 samples from the posterior distribution $p(\mu|X = 2)$. You are not allowed to use a Metropolis-Hastings sampler. You can assume you have access to random number generators for normal, gamma, binomial (including Bernoulli), Poisson, uniform, exponential, beta, and t distributions. If your algorithm simulates samples from a certain distribution, you must specify the exact values for the parameters, and you must provide all derivations of distributions in full.
[Hint: $p(\mu|X = 2) = p(\mu, \gamma = 1|X = 2) + p(\mu, \gamma = 0|X = 2)$.]

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**End of Exam — Total Available Marks = 50**

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